

# EC-467 Mobile Application Development



## BS (CE)-2021 LAB REPORT #4

**Submitted By:**

| Reg. No.  | Name            |
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|                 | Presentation<br>(Format(0.5)+<br>Title (0.5)+<br>Conclusion(1)<br>+Procedure(2)<br>+Objective(1)) | Calculation/<br>Coding | Observation/<br>Program Results | Total     |
|-----------------|---------------------------------------------------------------------------------------------------|------------------------|---------------------------------|-----------|
| <b>Total</b>    | <b>5</b>                                                                                          | <b>5</b>               | <b>5</b>                        | <b>15</b> |
| <b>Obtained</b> |                                                                                                   |                        |                                 |           |

**EC-467 Mobile Application Development**



## **BS (CE)-2021 LAB REPORT #5**

**Submitted By:**

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|                 | <b>Presentation<br/>(Format(0.5)+<br/>Title (0.5)+<br/>Conclusion(1)<br/>+Procedure(2)<br/>+Objective(1))</b> | <b>Calculation/<br/>Coding</b> | <b>Observation/<br/>Program Results</b> | <b>Total</b> |
|-----------------|---------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------|--------------|
| <b>Total</b>    | <b>5</b>                                                                                                      | <b>5</b>                       | <b>5</b>                                | <b>15</b>    |
| <b>Obtained</b> |                                                                                                               |                                |                                         |              |

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**Lab Instructor.**

**Dr. Muhammad Bilal**

# LAB REPORT #5

## List View and Array Adapter

### Introduction:

Android ListView is a view which groups several items and display them in vertical scrollable list. The list items are automatically inserted to the list using an Adapter that pulls content from a source such as an array or database. Android provides several subclasses of Adapter that are useful for retrieving different kinds of data and building views for an AdapterView ( i.e. ListView or GridView).

The common adapters are

- ArrayAdapter
- BaseAdapter

We can use this adapter when your data source is an array By default, ArrayAdapter creates a view for each array item by calling toString() on each item and placing the contents in a TextView. Consider you have an array of strings you want to display in a ListView, initialize a new ArrayAdapter using a constructor to specify the layout for each string and the string array.

```
ArrayAdapter adapter = new ArrayAdapter(this, R.layout.history_item_layout, countryArray);
```

### Class Task:

Country List by using List View and Array Adapter.

#### XML CODE OF MAIN ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <TextView
        android:id="@+id/list_tv"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Country List" />
```

```
<ListView
    android:id="@+id/country_list"
    android:layout_height="match_parent"
    android:layout_width="match_parent">
</ListView>
</LinearLayout>
```

## JAVA CODE OF MAIN ACTIVITY:

```
package com.example.madlab5;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ListView countryList=findViewById(R.id.country_list);
        ArrayList<String>countryArray=new ArrayList<String>();
        countryArray.add("Pakistan");
        countryArray.add("Saudi Arabia");
        countryArray.add("China");
        ArrayAdapter adapter=new ArrayAdapter(this,
            R.layout.activity_main,countryArray);
        countryList.setAdapter(adapter);
    }
}
```

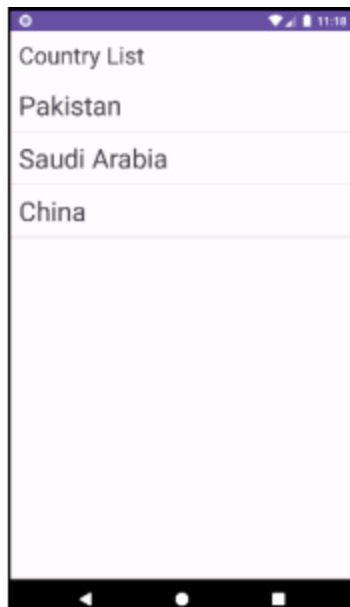
## Results:

---

```
2024-03-14 11:55:36: Launching app on 'Pixel 5 API 34'.
Starting: Intent { act=android.intent.action.MAIN cat=[android.intent.category.LAUNCHER] cmp=com.huzaifashahbaz.intentapp/.MainActivity }

Open logcat panel for emulator Pixel 5 API 34
Connected to process 2529 on device 'Pixel_5_API_34 [emulator-5554]'.
```

## OUTPUT:



## Home Task:

Consider your assignment # 1 and implement the same task using list view on history activity and use array list to add count in main activity and pass this array list to history activity.

Hint: ArrayList historyAL =

getIntent().getStringArrayListExtra(TasbeeActivity.MAIN\_TO\_HISTORY\_STRING\_KEY)

## XML CODE OF MAIN ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView"
        android:layout_width="131dp"
        </ android:layout_height="wrap_content"
```

```

        android:text="TextView"
        android:textColorLink="#805F5F"
        tools:layout_editor_absoluteX="267dp"
        tools:layout_editor_absoluteY="142dp" />
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
    <Button
        android:id="@+id/btn_count_1"
        android:layout_width="245dp"
        android:layout_height="wrap_content"
        android:onClick="onPressCountBtn"
        android:text="Count"
        tools:layout_editor_absoluteX="198dp"
        tools:layout_editor_absoluteY="156dp" />
    <Button
        android:id="@+id/btn_Rest_1"
        android:layout_width="142dp"
        android:layout_height="wrap_content"
        android:onClick="onPressResetBtn"
        android:text="Reset"
        tools:layout_editor_absoluteX="247dp"
        tools:layout_editor_absoluteY="224dp" />
</LinearLayout>
<Button
    android:id="@+id/btnViewhistory"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:onClick="onPressHistorybtn"
    android:text="View history"
    tools:layout_editor_absoluteX="296dp"
    tools:layout_editor_absoluteY="153dp" />
</LinearLayout>

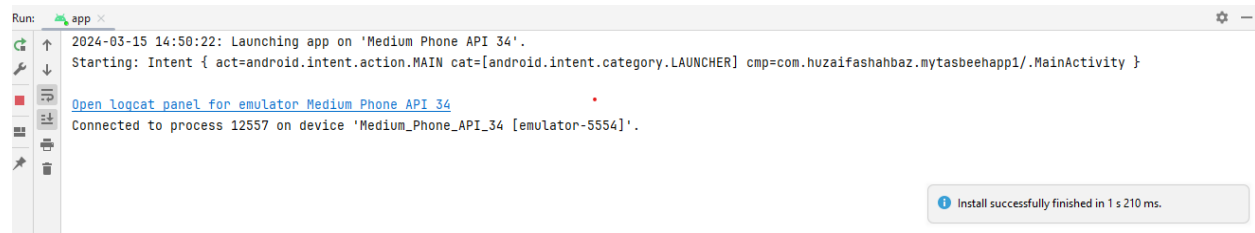
```

### **JAVA CODE OF MAIN ACTIVITY:**

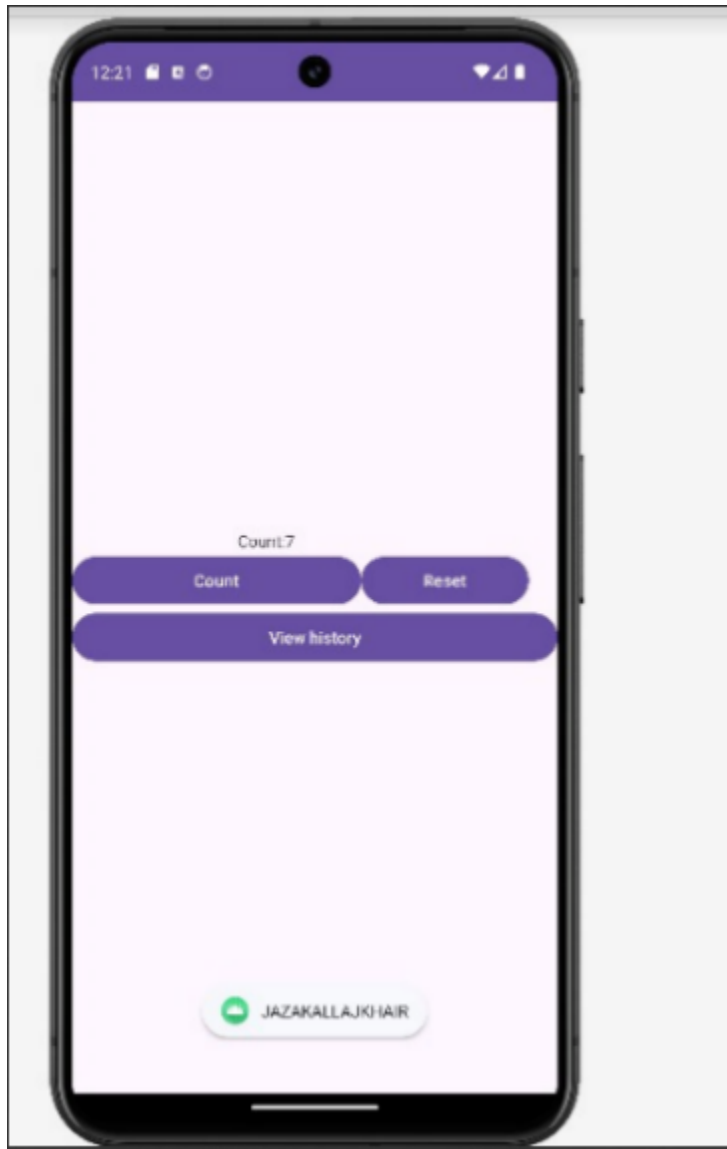
```
package com.huzaifashahbaz.mytasbeehapp1;
```

```
import android.os.Bundle;
import android.os.PersistableBundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity {
    TextView tvCount;
    Button btnCount,btnReset,btnViewHistory;
    int count=0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate( savedInstanceState );
        setContentView( R.layout.activity_main );
        tvCount=findViewById(R.id.textView);
    }
    public void onPressResetBtn(View View){
        count=0;
        tvCount.setText("Count:"+count);
    }
    public void onPressHistorybtn(View view){
        Toast.makeText( this, "JAZAKALLAJKHAIR", Toast.LENGTH_SHORT
).show();
    }
    public void onPressCountBtn(View View){
        count++;
        tvCount.setText("Count:"+count);
    }
}
```

## Results:



## OUTPUT:



## XML CODE OF HISTORY ACTIVITY:



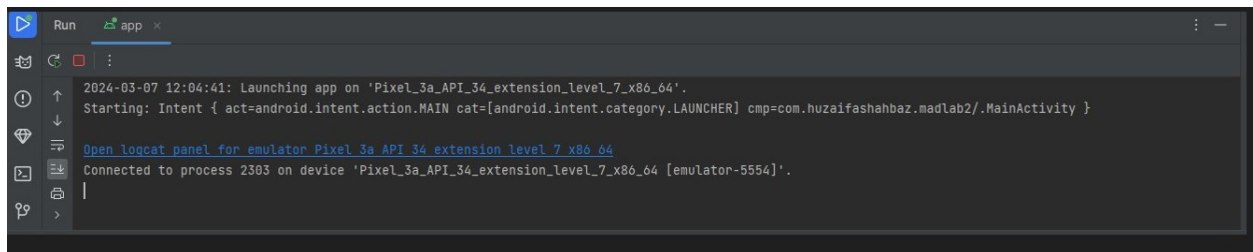
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/textView"
        android:layout_width="131dp"
        android:layout_height="wrap_content"
        android:text="TextView"
        android:textColorLink="#805F5F"
        tools:layout_editor_absoluteX="267dp"
        tools:layout_editor_absoluteY="142dp" />
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button
            android:id="@+id/btnViewhistory"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:onClick="onPressHistorybtn"
            android:text="View history"
            tools:layout_editor_absoluteX="296dp"
            tools:layout_editor_absoluteY="153dp" />
    </LinearLayout>

    <Button
        android:id="@+id/btnIncrement"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:onClick="onPressIncrementbtn"
        android:text="Increment"
        tools:layout_editor_absoluteX="296dp"
        tools:layout_editor_absoluteY="153dp" />
</LinearLayout>
```

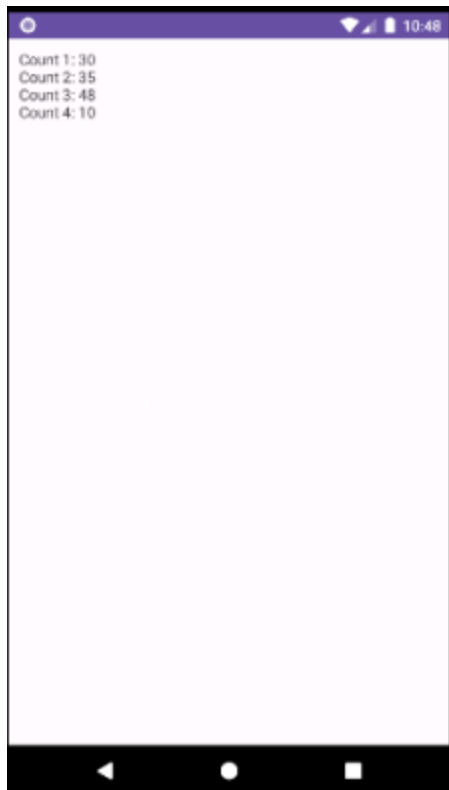
## JAVA CODE OF ACTIVITY HISTORY:

```
package com.huzaifashahbaz.assginmentoftabseehcounter;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class historyActivity extends AppCompatActivity {
    TextView tvCount;
    Button btnIncrement,btnViewHistory;
    int count=0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate( savedInstanceState );
        EdgeToEdge.enable( this );
        setContentView( R.layout.activity_history );
        tvCount=findViewById( R.id.textView );
    }
    void onPressHistorybtn(View view){
        count=0;
        tvCount.setText( "Count:"+count );
    }
    void onPressIncrementbtn(View view){
        count++;
        tvCount.setText("Count:"+count);
    }
}
```

## RESULTS:



## OUTPUT:



## XML CODE OF TASBEEHARRAY ACTIVITY:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=". TASBEEHARRAY Activity">
    <TextView
        android:id="@+id/historytext"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text=""
```

```

        android:textSize="20sp"
        android:textStyle="bold"
        android:fontFamily="causal"/>
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

</LinearLayout>

```

## JAVA CODE OF TASBEEHARRAY ACTIVITY:

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    countText = findViewById(R.id.countText);
    Button incrementButton = findViewById(R.id.incrementButton);
    Button historyButton = findViewById(R.id.historyButton);
    Button resetButton = findViewById(R.id.resetButton);

    incrementButton.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            count++;
            countText.setText("Count: " + count);
        }
    });
}

```

```

package com.huzaifashahbaz.assignmentoftabseehcounter;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class TASBEEHARRAYActivity extends AppCompatActivity {
    TextView tvCount;
    Button btnIncrement,btnViewHistory;
    int count=0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {

```

```

        super.onCreate( savedInstanceState );
        setContentView( R.layout.activity_tasbeeharray );
        countText=findbyViewById(R.id.countText);
        Button incrementButton=findbyViewById(R.id. incrementButton);
        Button historyButton=findbyViewById(R.id. historyButton);
        Button resetButton=findbyViewById(R.id. resetButton);
        incrementButton.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                count++;
                CountText.setText("Count:"+count);
            }
        });
        historyButton.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                Intent intent=new Intent(tasbeeharrayactivity,this,historyactivity.class);
                Intent.putIntegerArrayListExtra("Counthistory", historyactivity);
                StartActivity(intent);
            }
        });
        resetButton.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                CountHistory.add(count);
                Count=0;
                countText.setText("Count:0");
            }
        });

    }

    void onPressHistorybtn(View view){
        count=0;
    }

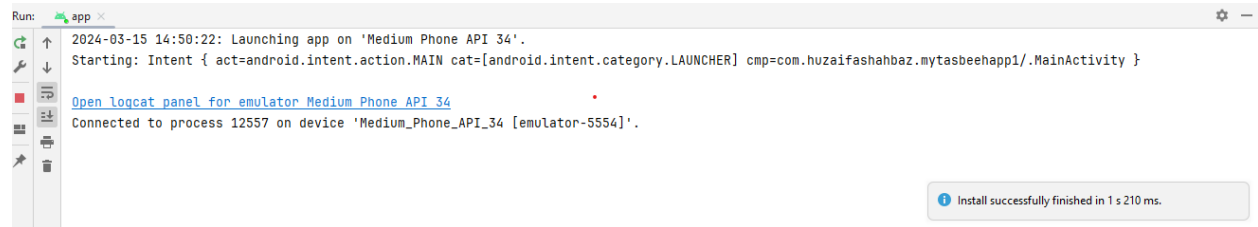
```

```

        tvCount.setText( "Count:"+count );
    }
    void onPressIncrementbtn(View view){
        count++;
        tvCount.setText("Count:"+count);
    }
}

```

## RESULTS:



## OUTPUT:



**Conclusion:**

In conclusion ,we had know about the functions and working of List View and Array Adapter.We also make the tasbeeh counter app by using list view and array adapter in mobile application development by using java code and xml linear layout code.